

AWS DeepRacer FAQs

General

Q: What is AWS DeepRacer

AWS DeepRacer is the fastest way to get rolling with reinforcement learning (RL), literally, with a fully autonomous 1/18th scale race car driven by reinforcement learning, 3D racing simulator, and a global racing league. Developers can train, evaluate, and tune RL models in the online simulator, deploy their models onto AWS DeepRacer for a real-world autonomous experience and compete in the AWS DeepRacer League for a chance to win the AWS DeepRacer Championship Cup. [AWS DeepRacer Device Terms, Warranties, and Notices »](#)

Q: What are the new features of AWS DeepRacer?

AWS DeepRacer customers can take on their next machine learning challenge using AWS DeepRacer with the launch of multi-car racing and object avoidance capabilities in the AWS DeepRacer console. Customers can now build models for object avoidance and dual-car head-to-head races by experimenting with multiple sensor inputs and the latest reinforcement learning algorithms and neural network configurations.

Q: How is AWS DeepRacer different from other robotic scale cars in the market?

AWS DeepRacer is the first autonomous scale car specifically developed to help developers get hands-on with reinforcement learning. AWS DeepRacer gives developers a simple way to learn RL, experiment with new RL algorithms and simulation-to-real domain transfer methods, and experience RL in the real world.

Q: What is the relationship between AWS DeepRacer and AWS Robocar Rally?

After the AWS DeepRacer launch, all Robocar Rally events will be replaced with AWS DeepRacer events. Robocar Rally inspired AWS DeepRacer, but unlike Robocar Rally, AWS DeepRacer focuses on RL and not on behavioral cloning.

Q: How can I get an AWS DeepRacer?

You can access the [AWS DeepRacer 3D racing simulator](#) in the AWS DeepRacer console to train your models, evaluate them, and take part in the AWS DeepRacer League. The AWS DeepRacer car is available for purchase on on [Amazon.com](#).

Q: How can I get an AWS DeepRacer Evo?

The AWS DeepRacer, AWS DeepRacer Evo, and AWS DeepRacer Evo sensor kit are available to order on Amazon.com for shipping within, as well as outside of, the USA (excluding the United Kingdom, Israel, Kazakhstan, Russia, and countries subject to OFAC sanctions). If you would like to be notified when it is available in your region, you can sign up to the [AWS DeepRacer Evo interest](#) list.

Q: Which geographic regions is AWS DeepRacer available in?

AWS Customers can access the AWS DeepRacer simulator from the US East (N. Virginia) Region.

Q: What is the AWS DeepRacer League?

The AWS DeepRacer League is the world's first global autonomous racing league for developers. AWS Customers can use their AWS DeepRacer RL models to compete in a global championship, racing for prizes, glory, and a chance to lift the Championship Cup.

Learn more about the league in the [dedicated FAQ section below](#)

Getting Started

Q: How do I get started with AWS DeepRacer simulator?

The [AWS DeepRacer simulator](#) provides a tutorial to get you started with reinforcement learning and training your first model. You will then also be able to evaluate and tune your models, before racing them in the AWS DeepRacer League. The AWS DeepRacer Developer Documentation provides additional details on [building your first model](#) and also [how to improve your models](#).

Q: How can customers get started with multi-car racing and object avoidance?

Customers can login into the [AWS DeepRacer console](#) from anywhere in the world. Here they can experiment with new sensor configurations in the Garage section of the console and build reinforcement learning models for head to head and object avoidance racing.

Q: How can I start my own Community Race?

Customers can create their own virtual races the Community Races section of the [AWS DeepRacer console](#).

Q: Do I need an AWS DeepRacer car in order to use the AWS DeepRacer simulator?

No. You can train models, evaluate them without owning an AWS DeepRacer car. Furthermore, you can race your models in both the AWS DeepRacer League Virtual and Summit Circuits without owning an AWS DeepRacer car.

Q: Does my DeepRacer need to be connected to the internet to race autonomously?

No. AWS DeepRacer uses the deployed RL model and input from the camera to run inference locally. AWS DeepRacer must be connected to the same Wi-Fi network as the device used to start and stop autonomous driving. Details on how to set up your vehicle can be found in the [Developer Documentation](#).

AWS DeepRacer League

Q: How do I race in the DeepRacer League?

Join the AWS DeepRacer League from anywhere in the world via the AWS DeepRacer console and put your skills to the test on virtual tracks monthly. Race for prizes and glory and a chance to advance to compete in the AWS DeepRacer Championship Cup. New for 2021 Developers will start in the Open Division of the league. Each month the top 10% in the Open Division will advance to the Pro Division the following month. The top 16 racers at the end of the monthly Pro Division race qualify for the Pro Finale, where they will race live to determine the winners for that month. The top 10 in the Pro Finale receive AWS DeepRacer Evo devices, while the top 3 will be eligible to receive an expenses-paid trip to advance to compete in the AWS DeepRacer Championship Cup at re:Invent.

Q: How many times can I enter the League?

There is no limit on the number of races or race formats you can enter, in-person or virtually. In fact, participating in multiple racing events, increases your chances of winning one of the prizes to advance to compete for the Championship Cup.

Q: What are the prizes?

Details of the AWS DeepRacer League prizes can be found [here](#). For full details please read the [AWS DeepRacer League official rules](#).

Q: Do I have to build my own model to race?

No, you do not need to build your own model to compete in the AWS DeepRacer League. There are tutorials inside the AWS DeepRacer simulator on how to build your own model, as well as pre-trained sample models you can use to get started.

Q: How long does a Virtual Circuit race last?

Each virtual race will last one month (from the 1st business day of the month to the last day of the month).

Q: How many Virtual Circuit races are there?

Starting on March 1st, the Virtual Circuit will have 8 monthly races, with three racing formats in each (time trial, object avoidance, and head-to-head). Each race will be revealed in the console on the first business day of each month leading up to re:Invent 2021 (March – October).

Q: How do I score points?

In 2021, the AWS DeepRacer league will no longer track cumulative points. Winners will be awarded based on single race performance only.

Q: What is AWS Summit Online?

The AWS Summit Online is a free online event designed to bring the cloud computing community together to connect, collaborate, and learn about AWS. At a Summit Online event, you can learn how to choose the right database, modernize your data warehouse, and drive digital transformation with AI using AWS services.

AWS DeepRacer Student

Q: What is AWS DeepRacer Student?

AWS DeepRacer Student, presented by Intel, lets high school and college-enrolled students around the globe develop their ML skills in a fun, hands-on autonomous racing league exclusively for students with no AWS account required. Participants can leverage 20 hours of ML educational material, 10 hours of monthly model training compute resources, and 5 GB of model storage to train a RL model in a simulated 3D racing environment.

Q: When does the AWS DeepRacer Student League begin?

The official AWS DeepRacer Student League season, kicked off on March 1, 2022. Competitions run monthly within AWS DeepRacer Student until September 30, 2022. The top participants in each monthly race will qualify for the Global AWS DeepRacer Student League Championships in October, 2022.

Q: Who can participate in the AWS DeepRacer Student League?

Any student over the age of 16 and currently enrolled in high school or undergraduate higher education at an accredited institution globally may apply, subject to terms and conditions.

Q: Does it cost anything to compete in the AWS DeepRacer Student League?

No, AWS DeepRacer Student is free for students to learn, train, and participate in the AWS DeepRacer Student League. Students can compete monthly to win prizes such as AWS DeepRacer devices and funds applied towards postsecondary tuition.

Q: Where can I find more details regarding AWS DeepRacer Student?

Go to [AWS DeepRacer Student](#) for more details.

Q: Where can I find the official AWS DeepRacer Student League rules?

For full details, please read the [AWS DeepRacer Student League Official Rules](#).

Product Details

Q: What are the product specifications of the AWS DeepRacer device?

Car: 1/18th scale 4WD with monster truck chassis

CPU: Intel Atom™ Processor

Memory: 4GB RAM

Storage: 32GB (expandable)

Wi-Fi: 802.11ac

Camera: 4 MP camera with MJPEG

Software: Ubuntu OS 16.04.3 LTS, Intel® OpenVINO™ toolkit, ROS Kinetic

Drive battery: 7.4V/1100mAh lithium polymer

Compute battery: 13600mAh USB-C PD

Ports: 4x USB-A, 1x USB-C, 1x Micro-USB, 1x HDMI

Sensors: Integrated accelerometer and gyroscope

Additional sensors on AWS DeepRacer Evo are LiDAR and 1 stereo 4 MP camera with MJPEG.

Q: If I already own an AWS DeepRacer car will it be able to master multi-car racing and object avoidance?

Developers who already own a DeepRacer car can buy an easy-to-install sensor kit from Amazon.com to give their car the same capabilities as AWS DeepRacer Evo.

Q: What does the LIDAR and stereo camera do?

LIDAR stands for light detection and ranging. It provides a continuous light beam providing the reinforcement model with inputs about whether a car is fast approaching from behind. The stereo camera adds depth perception to allow the car to detect objects in the road and be more responsive to its environment. Combining these sensory inputs with advanced algorithms, and updated reward functions, developers can build models that will not only detect obstacles (including other cars), but will also decide when to overtake to beat the other car to the finish line.

Q: Which AWS services are integrated with AWS DeepRacer?

AWS DeepRacer integrates with Amazon SageMaker for reinforcement learning model training, AWS RoboMaker to provide the racing simulator, Amazon Kinesis Video Streams for video streaming of virtual simulation footage, Amazon S3 for model storage, and Amazon CloudWatch for log capture.

Q: How do I add new tracks to the AWS DeepRacer simulator?

Currently, developers cannot add additional tracks to the AWS DeepRacer simulator. AWS DeepRacer will release a number of new racing tracks throughout 2021.

Q: Can I train my models on the AWS DeepRacer device?

No. Training an RL model requires feedback regarding the outcome of actions taken by the model. This feedback loop exists in the AWS DeepRacer simulator, but not in the real-world.

Q: Can I train my models locally on my own machine as opposed to the AWS Cloud?

Currently AWS DeepRacer does not support local training.

Q: Can I train AWS DeepRacer RL models directly on Amazon SageMaker?

Yes. You can use the AWS DeepRacer Distributed Training SageMaker Notebook to create and train RL models. You will be able to deploy these models to your AWS DeepRacer manually, but won't yet be able to import them into the AWS DeepRacer console.